COMPUTER NETWORKS-
DIGITAL ASSIGNMENT 2

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(I) Introduction to Network Simulation:

- Network simulation is a methodological framework for modeling, analyzing, and evaluating communication network behaviors by means of software tools.
- Instead of relying on physical networks, which involve significant costs, time, and a lack of flexibility in modification, network simulation allows for a controlled environment for effectively investigating a wide range of network scenarios.
- Communication networks in recent times have become significantly complex due to the convergence of heterogeneous communication technologies, including wired, wireless, and low-power communication infrastructures.
- It is necessary for communication networks to be evaluated for performance in a wide range of scenarios for reliable, efficient, and scalable communication.
- Network simulation allows researchers and engineers to effectively evaluate the performance of communication networks in a wide range of scenarios.
- Network simulation has a number of advantages, including the ability to easily modify parameters in communication networks, such as bandwidth, delay, and packet size, for effectively evaluating communication network performance in a wide range of scenarios.

- The applications of network simulation include:
  - Communication protocols
  - Performance evaluation of network topologies
  - Optimization of resource allocation
  - Evaluation of wireless and mobile communication networks
- Network simulation is of critical importance in the development of communication systems in recent times due to its ability to provide a versatile and efficient testing environment for communication networks.

(II) Overview of ns-3

- ns-3 is a discrete event simulation tool mainly used in research and teaching environments.
- It is used to simulate network protocols, devices, and communication systems in a realistic way.
- ns-3 is primarily implemented in C++. Python is also used, but this is optional. It is used in both academic and commercial environments to simulate networks.

Key Features of ns-3

- Discrete event simulation: In this simulation tool, events are processed in chronological order.

- Realistic protocol stack: It is used to simulate realistic network protocols such as IPv4, TCP, UDP, etc.

- Modular structure: It is divided into modules to accommodate different technologies.

- Supports different network technologies: It is used to simulate point-to-point networks, Wi-Fi networks using 802.11, low-rate networks using 802.15.4, Bluetooth networks, etc.

- Performance analysis: It is used to analyze the performance of a network, i.e., throughput, delay, packet loss, etc.

Working Principle

ns-3 is a tool used to simulate networks. It simulates networks in the following way:

1) Creating nodes

2) Creating communication links between nodes

3) Creating protocol stacks

4) Assigning IP addresses to nodes

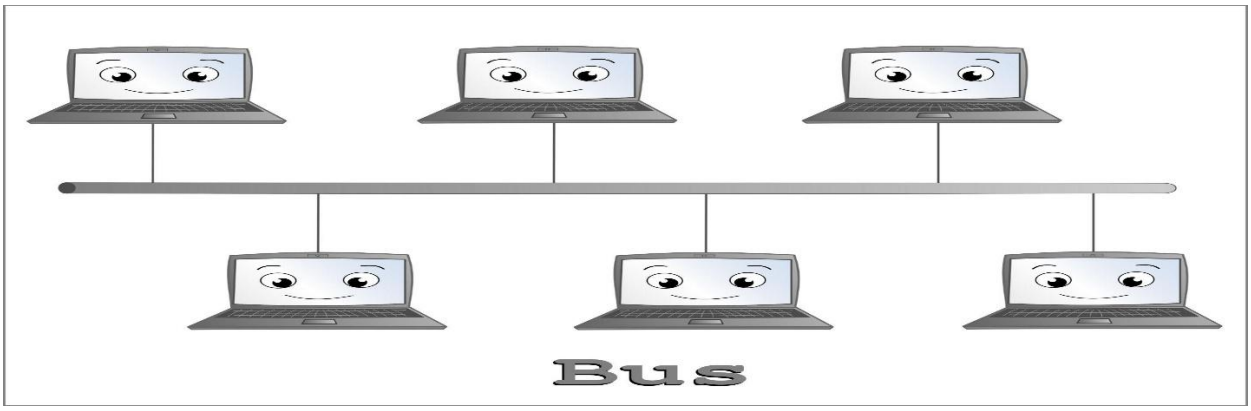
5) Creating applications

6) Running the simulation tool

ns-3 simulates the entire procedure of packet transmission and reception in a network. It simulates the transmission of a packet from a node to the destination node in a network.

(III) Network Topology Diagrams:

(a) Bus Topology:



In a bus topology, all nodes are connected to a single shared communication medium. Data transmitted by any node is available to all other nodes on the network.

Characteristics

- Single communication backbone
- Shared transmission medium
- Simple and cost-effective design

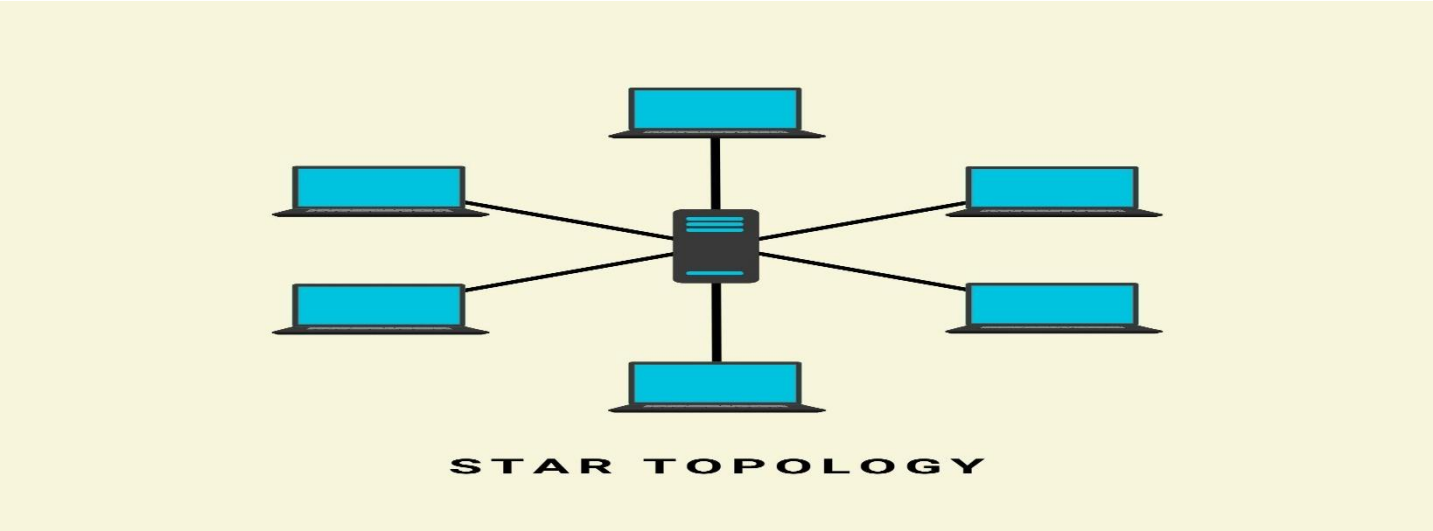
Advantages

- Easy to implement
- Requires less cabling

Disadvantages

- High chances of collisions
- Performance degrades as nodes increase
- Failure of backbone affects entire network

(b) Star Topology:



In a star topology, all nodes are connected to a central device such as a hub, switch, or access point.

Characteristics

- Centralized control
- Independent links between nodes and hub

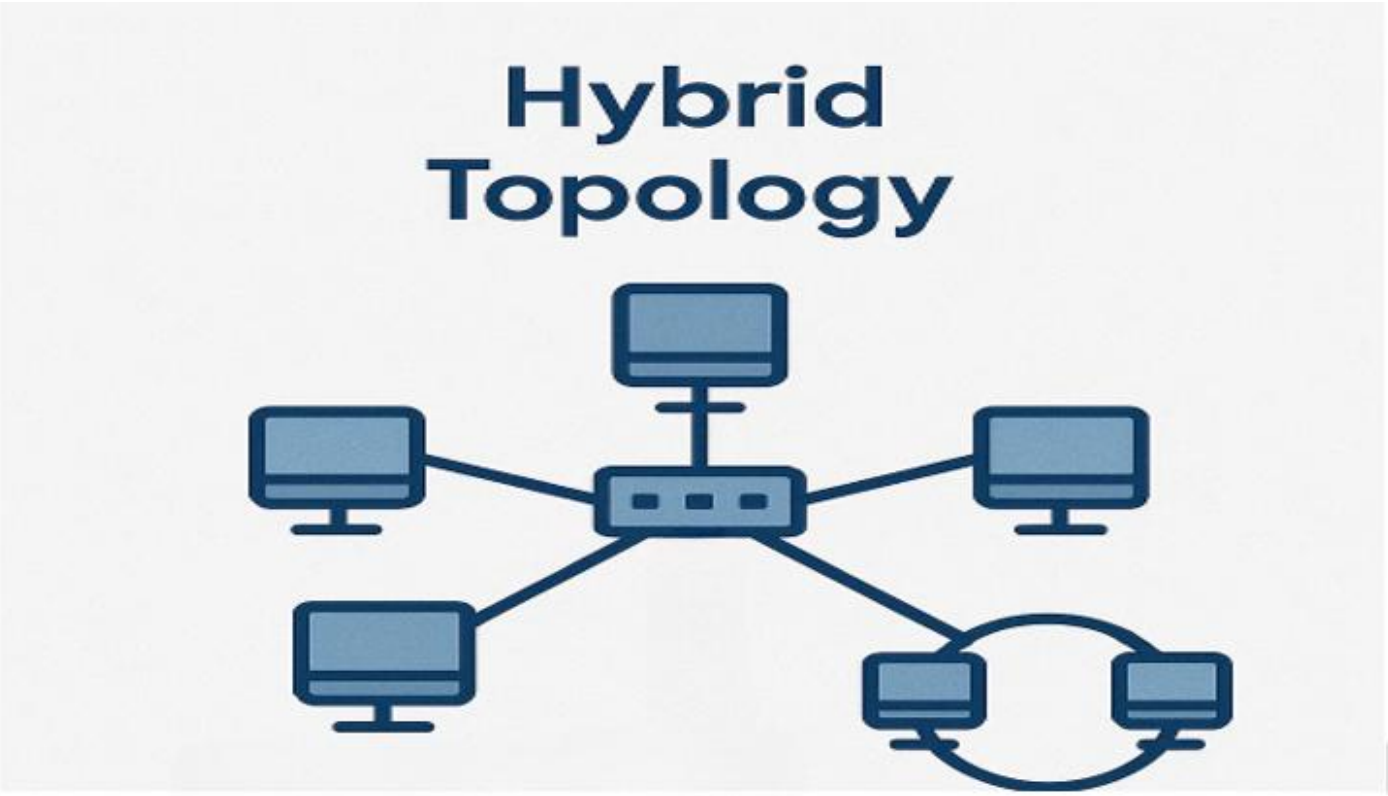
Advantages

- Easy fault detection
- Failure of one node does not affect others
- Better performance than bus topology

Disadvantages

- Central node is a single point of failure
- Higher cabling requirement

(c) Hybrid Topology (Consider Star+Ring):



Hybrid topology is a combination of two or more basic topologies such as star and ring.

Characteristics

- Flexible and scalable
- Combines advantages of multiple topologies

Advantages

- Improved reliability
- Customizable network structure

Disadvantages

- More complex design
- Higher implementation cost

(IV) Simulation Parameters:

Simulation parameters define the conditions under which the network operates. These parameters are crucial for ensuring consistent and meaningful performance analysis.

Parameter	Description
Number of Nodes	Total devices participating in the network
Simulation Time	Duration for which the simulation runs
Packet Size	Size of the transmitted data packets
Traffic Type	Type of application traffic (Eg: UDP Echo)
Data Rate	Rate of successful data delivery
Delay	Time taken for a packet to travel through the link
Channel Type	Wired or Wireless Medium
Throughput	Rate of successful data delivery
Packet Loss	Number of packets lost during the transmission
Jitter	Variation in packet delay

(V) Simulation Screenshots:

(a) Bus topology:

(i) Point-2-Point connection:

```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/bus-p2p
[0/2] Re-checking globbed directories...
[6/6] Linking CXX executable /home/keshav/ns-3-dev/build/scratch/ns3-dev-bus-wifi-default
[0/2] Re-checking globbed directories...
ninja: no work to do.

=== Bus P2P Metrics ===
Number of Nodes: 6
Simulation Time: 10 sec
Packet Size: 1024 bytes
Traffic Type: UDP Echo
Data Rate: 100 Mbps
Channel Type: Wired (CSMA)
Throughput: 9.39006 Kbps
Delay: 0.00346213 sec
Jitter: 0.00137653 sec
Packet Loss: 0
```

(ii) Wi-Fi connection

```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/bus-wifi
[0/2] Re-checking globbed directories...
ninja: no work to do.
[0/2] Re-checking globbed directories...
ninja: no work to do.

=== Bus WiFi Metrics ===
Number of Nodes: 6
Simulation Time: 10 sec
Packet Size: 1024 bytes
Traffic Type: UDP Echo
Data Rate: WiFi (High Rate)
Channel Type: Wireless
Throughput: 9.39254 Kbps
Delay: 0.002201 sec
Jitter: 0.001961 sec
Packet Loss: 0
```

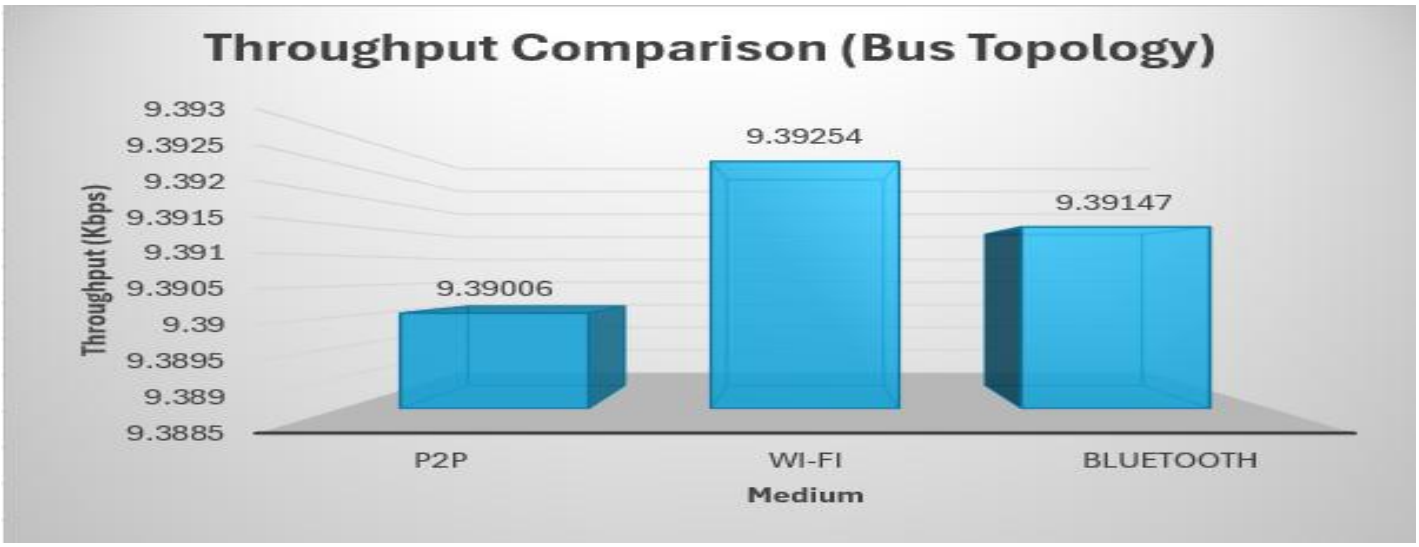
(iii) Bluetooth Connection

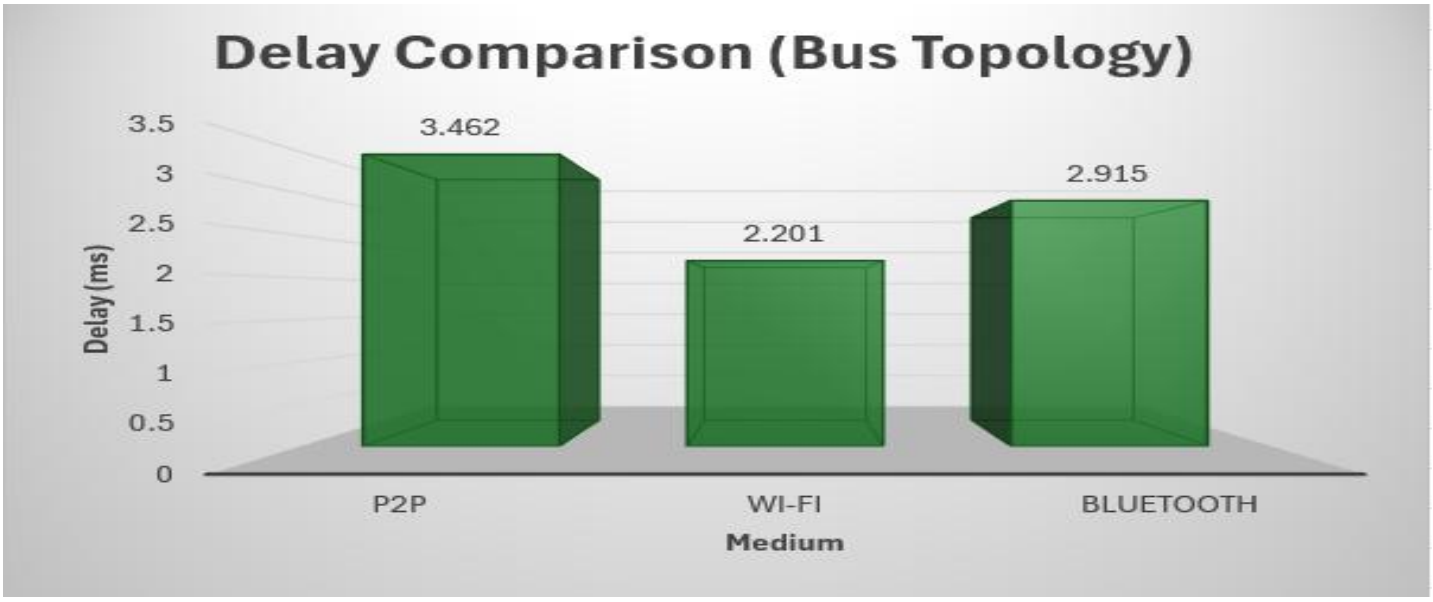
```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/bus-bt
[0/2] Re-checking globbed directories...
ninja: no work to do.
[0/2] Re-checking globbed directories...
ninja: no work to do.

=== Bus Bluetooth Metrics ===
Number of Nodes: 6
Simulation Time: 10 sec
Packet Size: 1024 bytes
Traffic Type: UDP Echo
Data Rate: Low (Bluetooth-like)
Channel Type: Wireless
Throughput: 9.39147 Kbps
Delay: 0.00291575 sec
Jitter: 0.00188175 sec
Packet Loss: 0
```

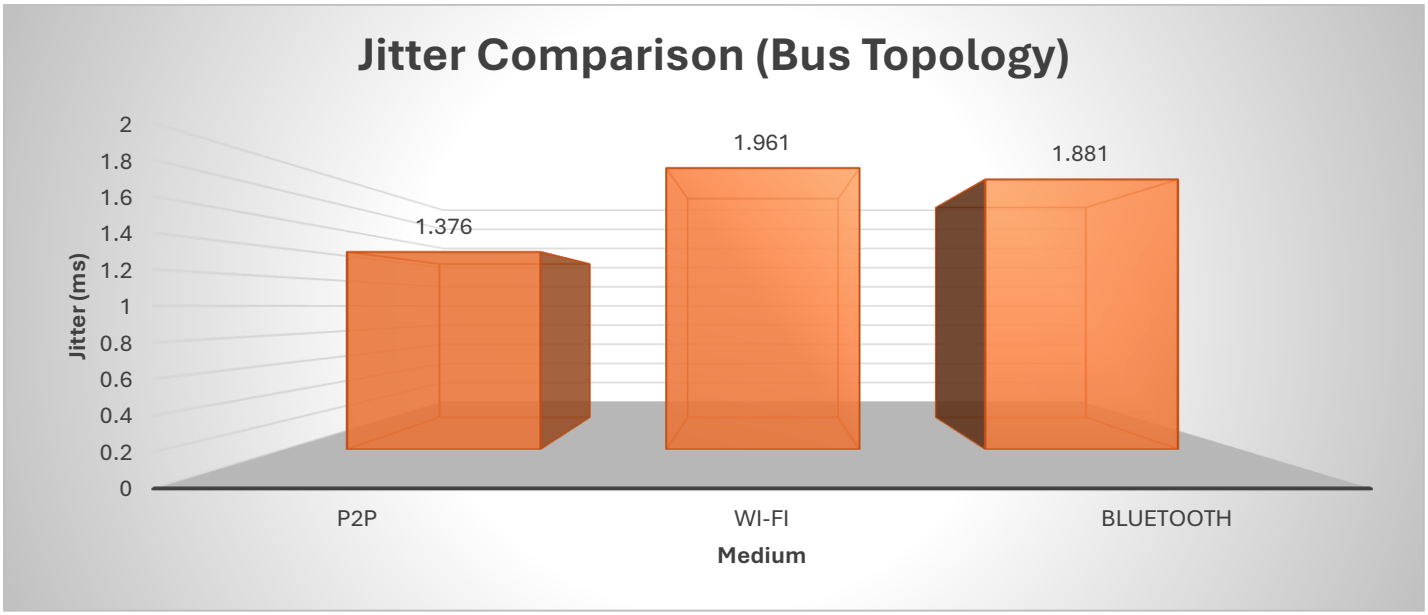
(iv) Comparison table:

Topology	Type of Connection	Throughput (Kbps)	Delay (ms)	Jitter (ms)	Packet Loss
Bus	Point-to-point	9.39006	3.462	1.376	0
Bus	Wi-Fi	9.39254	2.201	1.961	0
Bus	Bluetooth	9.39147	2.915	1.881	0

(v) Comparison Graphs:







vi) Bus Topology Analysis:

- The performance analysis of the bus topology for various media of communication shows that the throughput remains constant across all media, indicating that bandwidth limitations are not imposed by the network under the specified parameters of the simulation.
- However, the propagation delay varies significantly among all the media of communication. Wi-Fi offers the least delay, owing to its efficient mechanism of data transmission via a wireless medium. Bluetooth offers a moderate delay, consistent with its moderate data transmission speed. The highest delay is offered by the point-to-point CSMA configuration, owing to the shared nature of the medium.
- The analysis of the jitter of the network shows that Wi-Fi offers relatively stable data transmission compared to the other media of communication. Under the specified parameters of the simulation, no packet loss is reported by any of the media of communication.

(b) Star Topology:

(i) Point-2-Point Connection:

```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/star-p2p
[0/2] Re-checking globbed directories...
[4/4] Linking CXX executable /home/keshav/ns-3-dev/build/scratch/ns3-dev-star-wifi-default
[0/2] Re-checking globbed directories...
ninja: no work to do.

=== Star P2P Metrics ===
Number of Nodes: 7
Simulation Time: 10 sec
Packet Size: 1024 bytes
Traffic Type: UDP Echo
Data Rate: 100 Mbps
Channel Type: Wired (Hub-based)
Throughput: 9.39006 Kbps
Delay: 0.00308713 sec
Jitter: 0.00100153 sec
Packet Loss: 0
```

(ii) Wi-Fi Connection:

```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/star-wifi
[0/2] Re-checking globbed directories...
ninja: no work to do.
[0/2] Re-checking globbed directories...
ninja: no work to do.

=== Star WiFi Metrics ===
Number of Nodes: 7
Simulation Time: 10 sec
Packet Size: 1024 bytes
Traffic Type: UDP Echo
Data Rate: WiFi (High Rate)
Channel Type: Wireless
Throughput: 9.39254 Kbps
Delay: 0.002451 sec
Jitter: 0.002211 sec
Packet Loss: 0
```

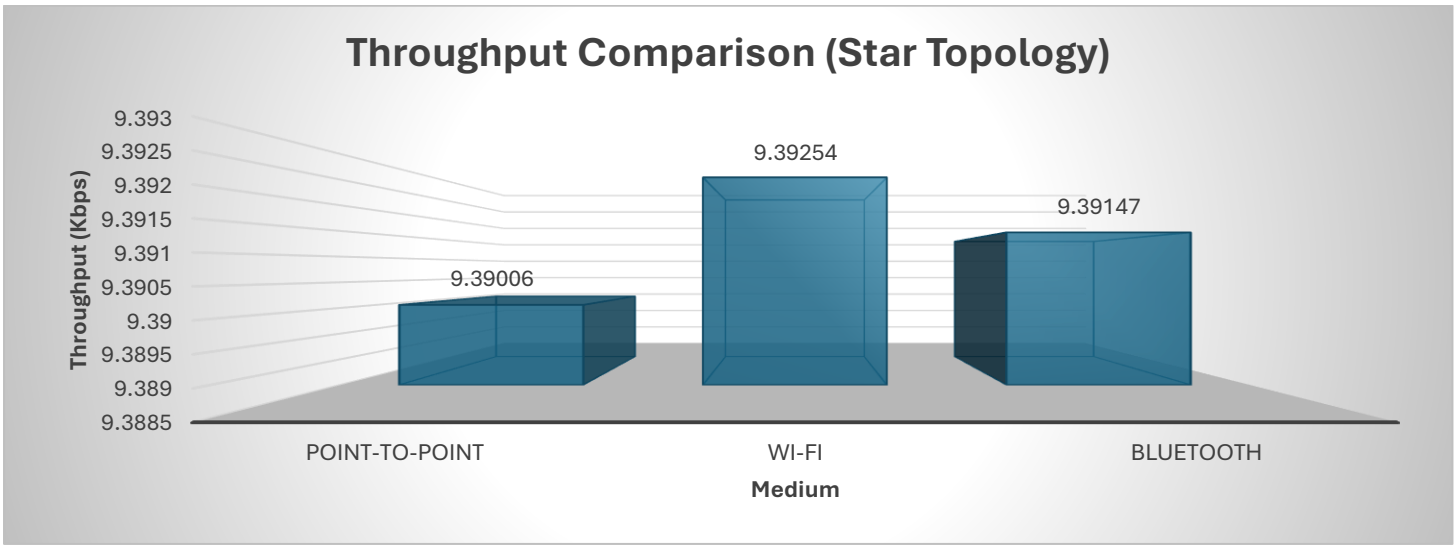
(iii) Bluetooth connection:

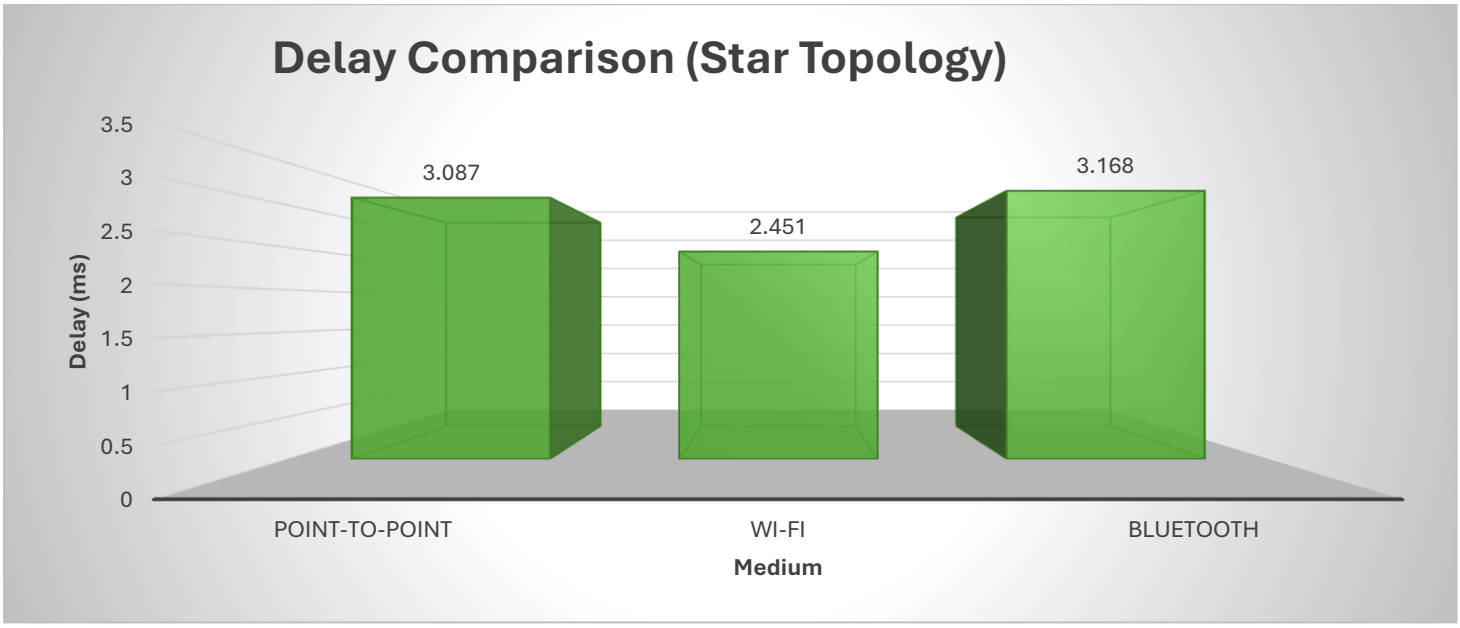
```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/star-bt
[0/2] Re-checking globbed directories...
ninja: no work to do.
[0/2] Re-checking globbed directories...
ninja: no work to do.

=== Star Bluetooth Metrics ===
Number of Nodes: 7
Simulation Time: 10 sec
Packet Size: 1024 bytes
Traffic Type: UDP Echo
Data Rate: Low (Bluetooth-like)
Channel Type: Wireless
Throughput: 9.39147 Kbps
Delay: 0.00316825 sec
Jitter: 0.00213425 sec
Packet Loss: 0
```

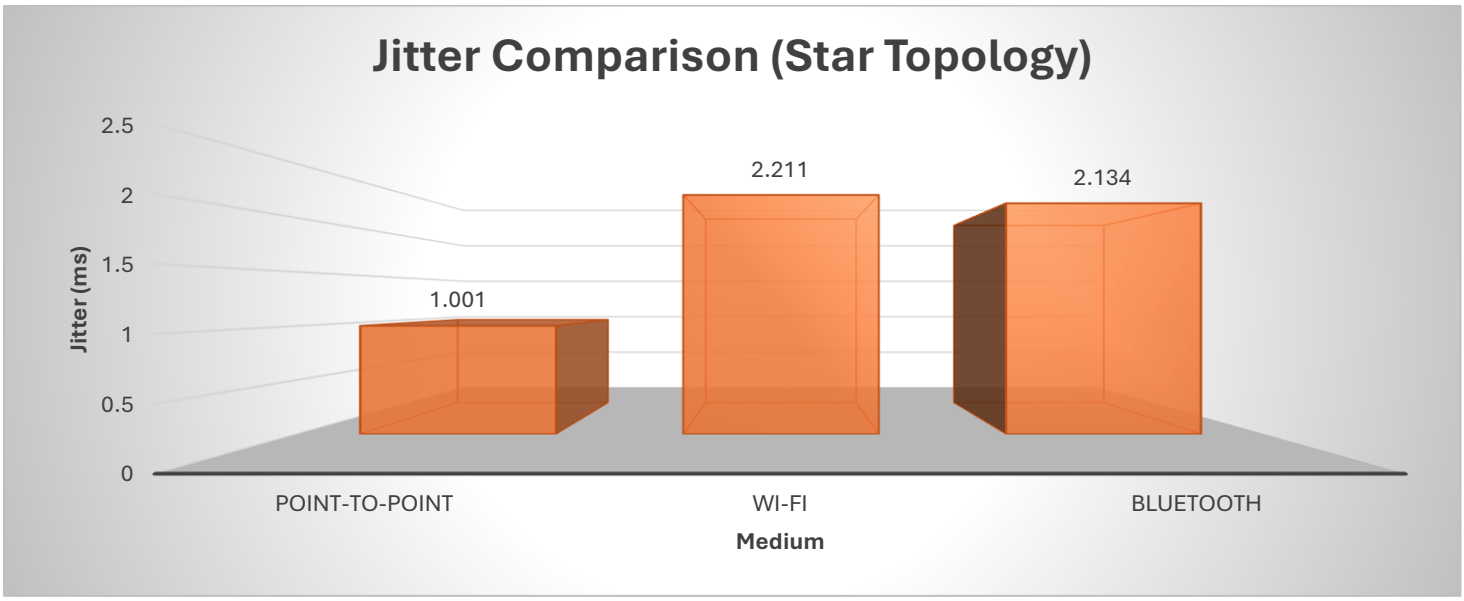
(iv) Comparison Table:

Topology	Type of Connection	Throughput (Kbps)	Delay (ms)	Jitter (ms)	Packet Loss
Star	Point-to-point	9.39006	3.087	1.001	0
Star	Wi-Fi	9.39254	2.451	2.211	0
Star	Bluetooth	9.39147	3.168	2.134	0

(v) Comparison Graphs







(vi) Star Topology Analysis:

- In the performance assessment of the star topology, the effect of the centralized hub is made clear. Throughput is uniform across all the mediums of communication. This shows the efficiency of the network without any bandwidth issues.
- In the delay analysis, the star topology shows the lowest delay with the use of Wi-Fi at 2.451 ms, followed by the point-to-point configuration with a slightly higher delay of 3.087 ms. However, the highest delay is observed with the use of the Bluetooth medium.
- In the jitter analysis, the highest stability is observed with the use of the point-to-point configuration with the lowest jitter value of 1.001 ms. However, the wireless mediums of communication show higher jitter due to the inherent nature of wireless communication.
- No packets are lost with any of the configurations, thus showing the reliability of the data transfer with the star topology.
- In conclusion, the star topology shows structured communication with a higher stability of the network.

(c) Hybrid Topology (Star+Ring)

(i) Point-2-Point Connection:

```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/hybrid-p2p
[0/2] Re-checking globbed directories...
[2/2] Linking CXX executable /home/keshav/ns-3-dev/build/scratch/ns3-dev-hybrid-p2p-default
[0/2] Re-checking globbed directories...
ninja: no work to do.

=== Hybrid P2P Metrics ===
Number of Nodes: 7
Simulation Time: 10 sec
Packet Size: 1024 bytes
Traffic Type: UDP Echo
Data Rate: 100 Mbps
Channel Type: Hybrid Wired
Throughput: 9.38447 Kbps
Delay: 8.51139 ms
Jitter: 2.25459 ms
Packet Loss: 0
```

(ii) Wi-Fi Connection:

```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/hybrid-wifi
[0/2] Re-checking globbed directories...
ninja: no work to do.
[0/2] Re-checking globbed directories...
ninja: no work to do.

=== Hybrid WiFi Metrics ===
Number of Nodes: 7
Simulation Time: 10 sec
Packet Size: 1024 bytes
Traffic Type: UDP Echo
Data Rate: WiFi (High Rate)
Channel Type: Hybrid Wireless
Throughput: 9.39254 Kbps
Delay: 2.451 ms
Jitter: 2.211 ms
Packet Loss: 0
```

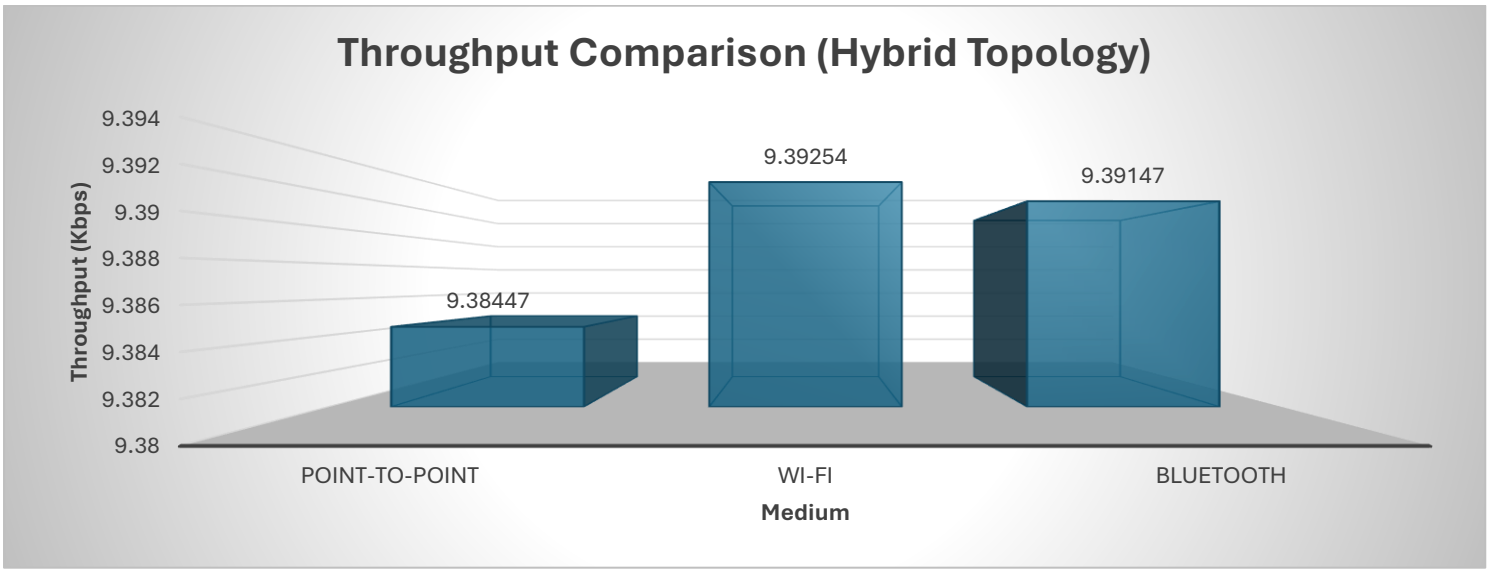
(iii) Bluetooth Connection:

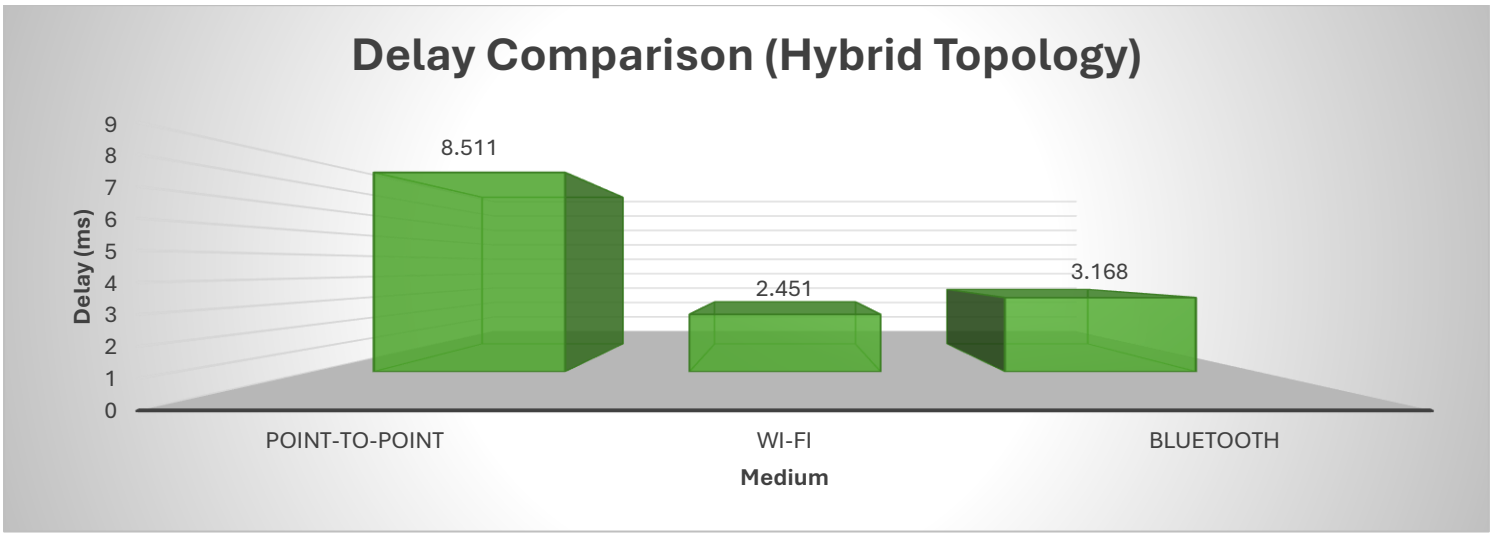
```
keshav@UbuntuOs:~/ns-3-dev$ ./ns3 run scratch/hybrid-bt
[0/2] Re-checking globbed directories...
ninja: no work to do.
[0/2] Re-checking globbed directories...
ninja: no work to do.

=== Hybrid Bluetooth Metrics ===
Number of Nodes: 7
Simulation Time: 10 sec
Packet Size: 1024 bytes
Traffic Type: UDP Echo
Data Rate: Low (Bluetooth-like)
Channel Type: Hybrid Wireless
Throughput: 9.39147 Kbps
Delay: 3.16825 ms
Jitter: 2.13425 ms
Packet Loss: 0
```

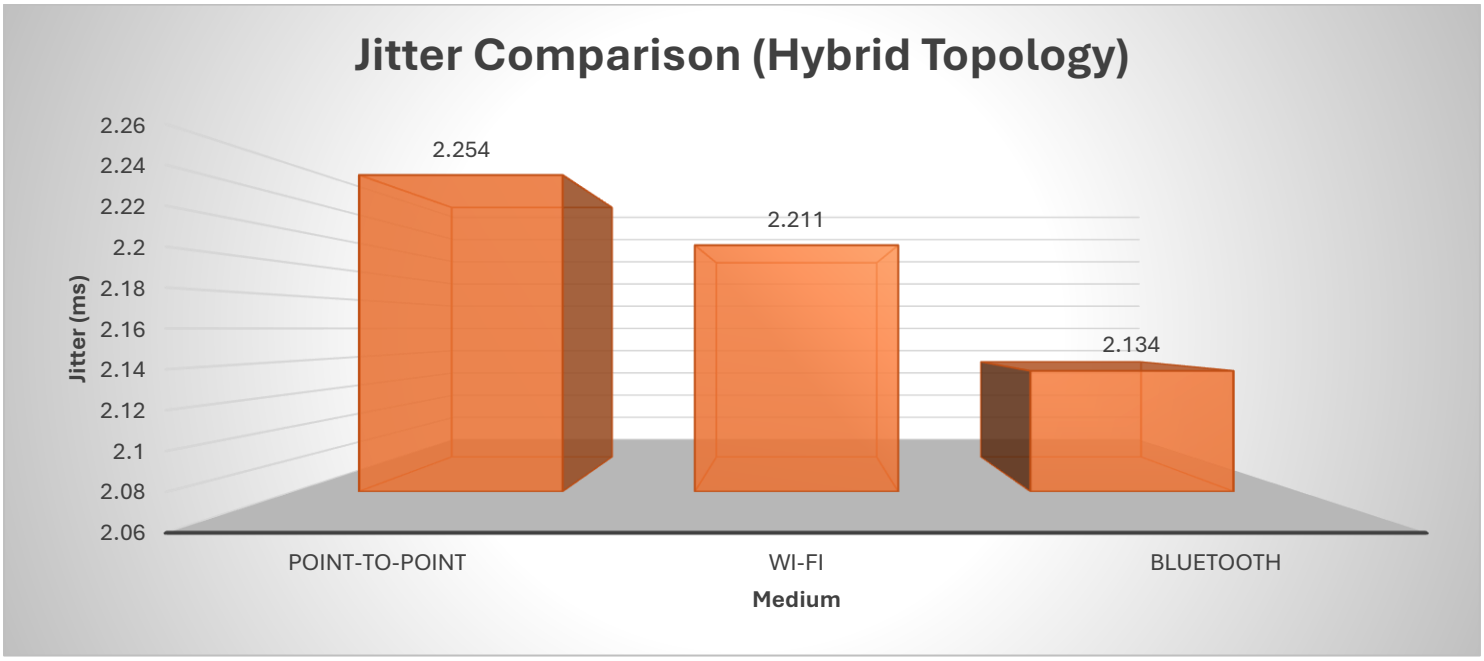
(iv) Comparison Table:

Topology	Type of Connection	Throughput (Kbps)	Delay (ms)	Jitter (ms)	Packet Loss
Hybrid	Point-to-point	9.38447	8.511	2.254	0
Hybrid	Wi-Fi	9.39254	2.451	2.211	0
Hybrid	Bluetooth	9.39147	3.168	2.134	0

(v) Comparison Graphs:







vi) Hybrid Topology Analysis:

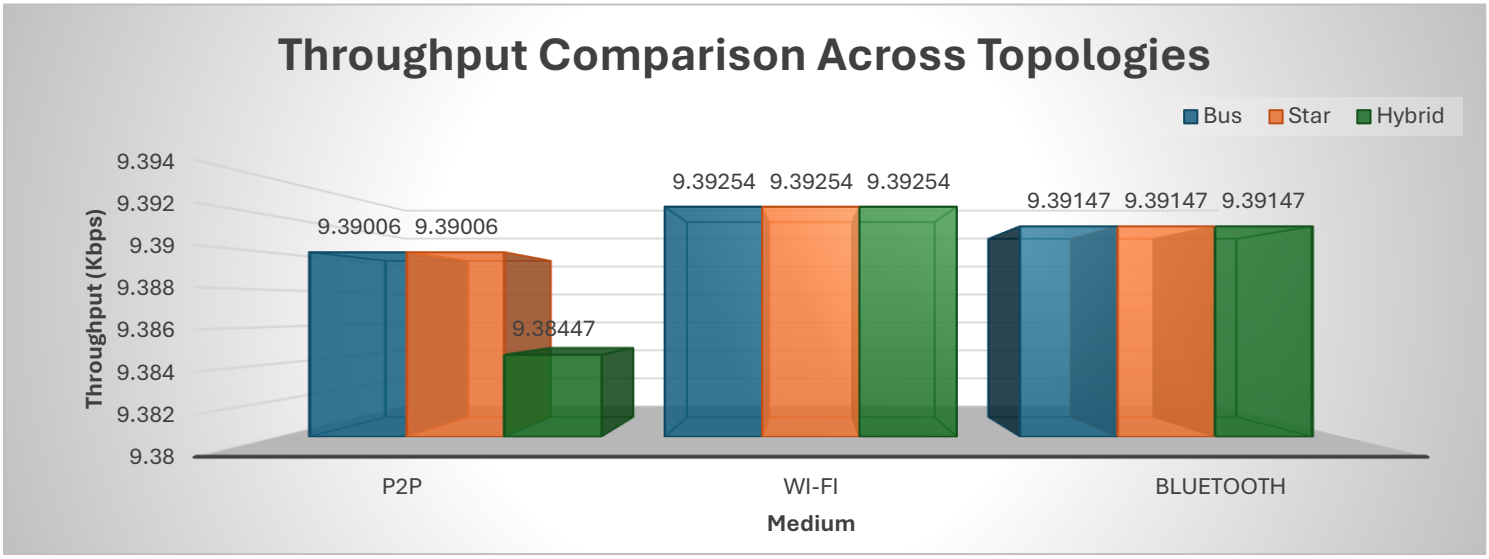
- The hybrid topology combines the characteristics of the star topology and the ring topology. This leads to a level of increased complexity in the routing of the network.
- In the different types of communication media, the throughput level remains the same. This shows that the management of the data transfer process is efficient, given the level of increased complexity in the network structure.
- In the analysis of the delays in the network, there is a marked increase in the point-to-point network, with the delays being as high as 8.511 ms. This is higher than the delays experienced in the bus topology and the star topology. This shows that the level of complexity in the network structure, given the nature of the communication, leads to the elevation of the delays. In the different types of network topologies, the lowest level of delay is experienced in the Wi-Fi network, with the delays being as low as 2.451 ms. In the Bluetooth network, the level of delay is moderate.
- In the analysis of the network jitter, it is evident that the level of jitter is higher in the network compared to the other types of network topology. In the wireless network, the level of jitter is higher. In the different types of network topology, there is no loss of the packets. This shows that the level of complexity in the network structure does not affect the efficiency of the data transfer process. In summary, the level of flexibility in the network topology is high; however, the level of delay is higher compared to the other types of network topology.

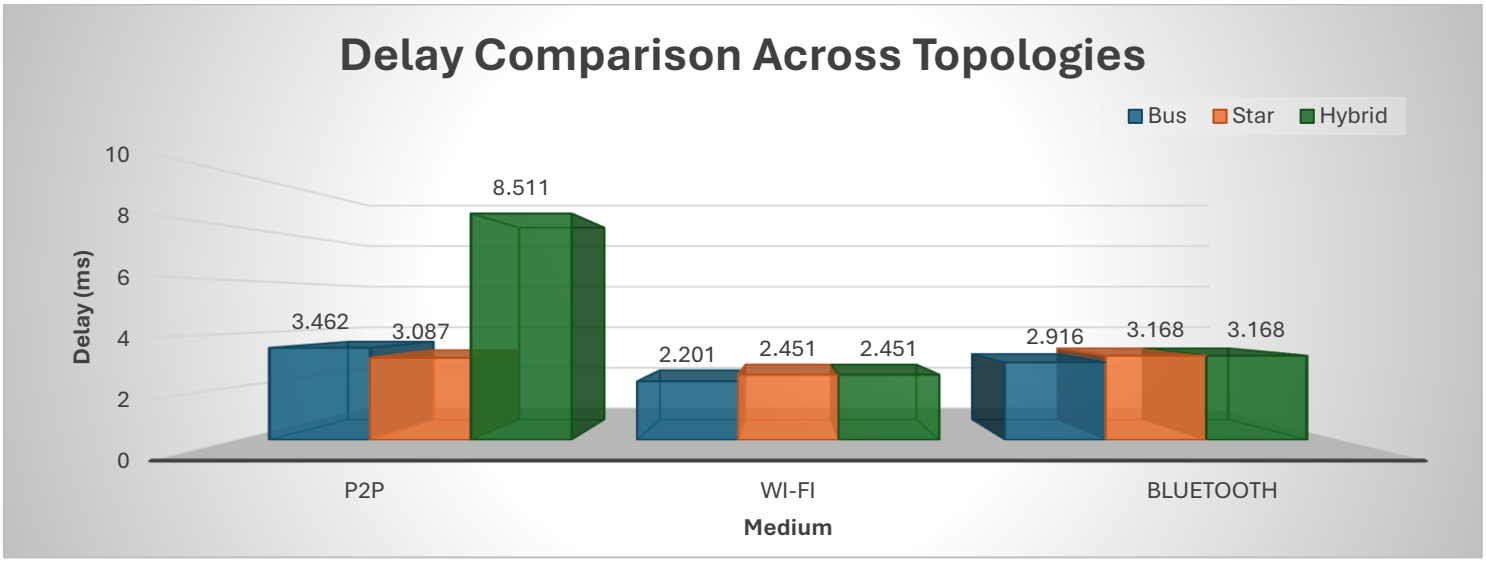
VI) FINAL RESULTS:

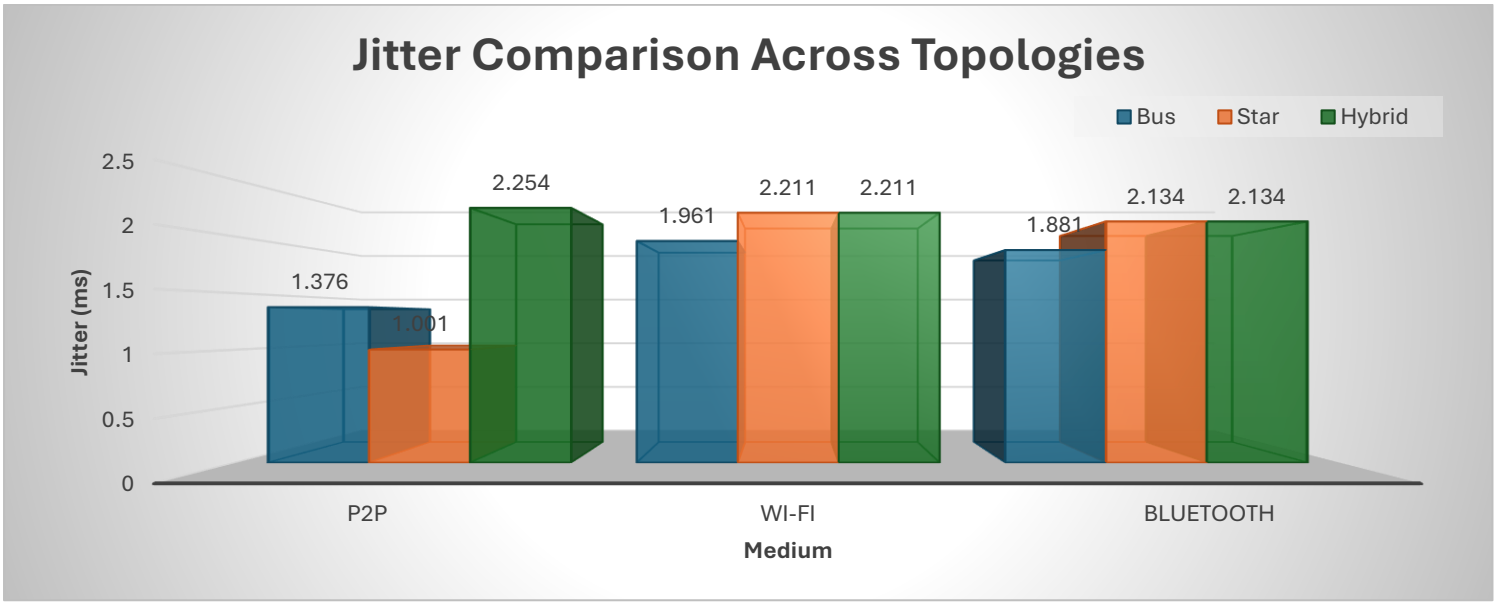
Comparison Table:

Topology	Medium	Throughput (Kbps)	Delay (ms)	Jitter (ms)	Packet Loss
Bus	P2P	9.39006	3.462	1.376	0
Bus	Wi-Fi	9.39254	2.201	1.961	0
Bus	Bluetooth	9.39147	2.916	1.881	0
Star	P2P	9.39006	3.087	1.001	0
Star	Wi-Fi	9.39254	2.451	2.211	0
Star	Bluetooth	9.39147	3.168	2.134	0
Hybrid	P2P	9.38447	8.511	2.254	0
Hybrid	Wi-Fi	9.39254	2.451	2.211	0
Hybrid	Bluetooth	9.39147	3.168	2.134	0

VII) COMPARISON GRAPHS







VIII) COMPARITIVE DISCUSSION:

- In a comparative evaluation of the performance of the bus, star, and hybrid topologies, it is evident that there is a significant variation in the characteristics of the topologies when used with different media of communication.
- Throughput is constant in all topologies, which indicates that the network topology has a minimal effect on the efficiency of data transmission in the network.
- In terms of delay, there is significant variation in the topologies, with the hybrid network showing the highest delay. This is more evident when using the point-to-point network configuration. This is possibly due to the increased routing complexity of the network. On the other hand, the bus and star topologies show lower values of delay, with the Wi-Fi configuration showing the lowest delay among the media of communication.
- Analysis of the jitter parameter shows stable communication with the bus and star topologies, while the hybrid network shows significant variation in the data transmission times. This is possibly due to the increased complexity of the network.
- On the other hand, no packet loss is observed with any of the network topologies, which indicates that data transmission is reliable with these network topologies and the parameter set used in the evaluation process.
- Although the hybrid network shows increased flexibility and efficiency, it shows relatively lower efficiency in terms of delay and jitter when compared with the star network, which shows increased efficiency in terms of several network parameters. On the other hand, the bus network shows simplicity at the expense of moderate efficiency.

IX) CONCLUSION

- From the performance analysis carried out on different network topologies, it is evident that the architecture of the network affects the delay and stability, whereas it affects the throughput to a very small extent.
- Simpler networks, like the bus and star networks, offer efficient and stable communication, whereas the hybrid networks cause a delay due to the increased complexity of the routing.
- Wireless networks, especially Wi-Fi, have been found to offer less delay and therefore are more appropriate for real-time applications.